Day 1: Gen-AI Landscape

Gen-AI + Agentic AI with JavaScript (Gemini Edition) | Phase 0: Setup

1. Concept

AI, ML, DL, LLM, Gen-AI: ye sab ek jaisa nahi hai

Ye ek **Russian doll (matryoshka)** ki tarah hai. Har bada concept apne andar chhota concept rakhta hai.

```
AI (Artificial Intelligence)
 |-- ML (Machine Learning)
      |-- DL (Deep Learning)
           |-- Gen-AI (naya content banata hai)
                |-- LLM (text samajhta aur likhta hai)
```

Term	English	Hinglish
AI	Any system that mimics human intelligence	Koi bhi machine jo insaan jaisa "sochne" ka kaam kare
ML	AI that learns patterns from data instead of hand-written rules	Rules khud likhne ki jagah, data se patterns seekhna
DL	ML using multi-layer neural networks	ML ka powerful version, jisme bahut saari layers hoti hain
Gen-AI	AI that *creates* new content (text, image, code, audio)	AI jo kuch **naya banata** hai, sirf predict ya classify nahi karta
LLM	A huge model trained on text that predicts the next word	Bahut bada model jo text pe train hua hai, jaise Gemini, GPT, Claude

Real-life analogy (recipe)

- **Traditional programming:** Tum recipe (rules) khud likhte ho. if onion hai, fry karo.
- **ML:** Tum 10,000 dishes dikhate ho, aur machine khud seekh leti hai ki accha khana kaisa hota hai.
- **Gen-AI / LLM:** Machine ne itni saari recipes padh li hain ki ab tum bolo "kuch spicy bana do" aur wo **nayi recipe likh deti hai**.

Gen-AI se pehle vs ab

Pehle (Traditional ML)	Ab (Gen-AI)
Ek task ke liye ek model (spam detector alag, translator alag)	Ek model, bahut saare tasks
Training data aur ML team chahiye	Sirf API call + achha prompt
Months lagte the	Ghanton me prototype

Isi wajah se developers ki demand badhi hai. Ab har product me AI feature lagana hai, aur model khud train karne ki jagah usse **integrate** karna hota hai. Ye kaam MERN developer already achhe se kar sakta hai.

Job roles

Role	Kya karta hai	Tumhare liye?
ML Engineer / Data Scientist	Models train karta hai, math aur stats heavy	Abhi nahi
AI Engineer	Existing LLMs ko use karke apps banata hai (RAG, agents, tools)	**Target**

Role	Kya karta hai	Tumhare liye?
Full-stack AI Developer	MERN + AI features	**Target**

Key insight: AI Engineer banne ke liye model train karna zaroori nahi. APIs, prompts, RAG, agents, aur production engineering ka skill chahiye. Python ki bhi zaroorat nahi hai.

2. Workflow

Ek typical Gen-AI feature MERN app me aise fit hota hai:

```
User (React UI)
     |  "Mera resume improve karo"
     v
Express Server (Node.js)      <- API key yahan safe rehti hai
     |  prompt + data
     v
LLM API (Gemini)
     |  generated response
     v
Express Server -> (MongoDB me save / process)
     |
     v
React UI (response dikhao)
```

Important rule: Gemini API key **kabhi React (frontend) me mat daalna**. Hamesha backend se call karo, warna key leak ho jayegi.

Aaj ka setup workflow

1. Node.js version check karo
2. Repo me Day 1 folder banao
3. npm init karo aur ES modules enable karo
4. .gitignore me .env daalo (aaj hi, taaki key galti se push na ho)
5. Neeche wala demo script run karo

3. Code

Setup (terminal me)

```
node -v        # 20 ya usse upar hona chahiye
cd your-repo
mkdir -p phase-0-setup/day-01 && cd phase-0-setup/day-01
npm init -y
```

package.json me ye line add karo taaki import syntax chale:

```
"type": "module"
```

Repo ke root me .gitignore check karo:

```
node_modules
.env
```

Demo: Rule-based vs AI ka farak

Aaj API call nahi karenge (wo Day 2 me). Aaj ye samjhenge ki **traditional code kahan fail hota hai**, aur LLM ki zaroorat kyun padi. File: **rule-based.js**

```
// Traditional approach: hum khud rules likhte hain
function ruleBasedSentiment(text) {
  const positive = ["good", "great", "love", "amazing"];
  const negative = ["bad", "worst", "hate", "terrible"];
```

```
  const t = text.toLowerCase();

  if (positive.some((w) => t.includes(w))) return "positive";
  if (negative.some((w) => t.includes(w))) return "negative";
  return "unknown";
}

const reviews = [
  "This phone is amazing, I love it",   // easy
  "The food was not good",              // negation
  "Yeh movie ekdum bakwaas thi",        // Hinglish
  "Great, another delayed flight...",   // sarcasm
];

for (const r of reviews) {
  console.log(`"${r}" -> ${ruleBasedSentiment(r)}`);
}
```

Run: `node rule-based.js`. Expected output:

```
"This phone is amazing, I love it"   -> positive   (sahi)
"The food was not good"              -> positive   (galat!)
"Yeh movie ekdum bakwaas thi"        -> unknown    (galat)
"Great, another delayed flight..."   -> positive   (sarcasm, galat)
```

Seekh: Rules se hum negation, Hinglish aur sarcasm kabhi poori tarah handle nahi kar sakte. LLM ye sab context se samajh leta hai, bina ek bhi rule likhe. Day 2 me hum yehi problem Gemini se solve karenge.

Homework (10 min)

1. Upar ka code **khud type karo**, copy-paste nahi.
2. 2 aur tricky reviews add karo jahan rule-based fail ho.
3. Google AI Studio (aistudio.google.com) kholo, API key ready rakho, aur Playground me kuch prompts try karo.
4. Code GitHub pe push karo.

4. Quick Revision (Quiz Answers)

Question	Answer
Sabse bada concept kaunsa hai?	AI (Artificial Intelligence)
Traditional ML vs Gen-AI ka sabse bada farak?	Gen-AI naya content banata hai, aur ek model bahut saare tasks kar sakta hai
AI Engineer banne ke liye sabse zaroori kya?	Existing LLMs ko APIs, RAG aur agents ke zariye apps me integrate karna
Gemini API key kahan rakhein?	Backend ki .env file me, aur .env ko .gitignore me daalo
Rule-based code ne "not good" ko positive kyun bataya?	Sirf keyword match hota hai, "not" ka context nahi samajhta

Next: Day 2: Repo setup, .env, API key safety, Google AI Studio tour, pehli API call with @google/genai